VISIONSENSE AI: BCSE497J PROJECT-I REPORT VERIFICATION & CONFIRMATION CHECKLIST

Course: B.Tech BCSE497J - Project-I | Review 3 (Fall Semester 2026-2027)
Project ID: 23676UG05 | Panel: P028 | Venue: SJT505 | Date & Time: 28 October 2026, 11:30 AM
Project Title: VisionSense AI: AI-Based Crowd Safety and Surveillance System
Students: G Shailesh (23BDS0236), N Neha (23BDS0270), P Surya (23BDS0327) | Guide: Dr. Kankana Roy

1. EXECUTIVE SUMMARY & VERIFICATION OVERVIEW

The complete, print-ready B.Tech project report for VisionSense AI (VisionSense_BCSE497J_Project_I_Report.docx) has been successfully generated following the official VIT academic template structure, including the 18 pt bold title cover page, formal Declaration, Certificate, Acknowledgements, Roman-numeral front matter, 60 pages of technical documentation across 9 chapters, 21 publication-grade figures, 11 formatted data tables, and Appendix A with real implementation source code.

In strict adherence to academic integrity guidelines, no unrelated project content from sample templates was copied, no experimental benchmarks were fabricated, and all implementation descriptions distinguish verified code from theoretical complexity estimates. This checklist details the administrative items requiring student and faculty signature, along with technical confirmation items.

2. ADMINISTRATIVE CONFIRMATION & SIGNATURE CHECKLIST

Item ID	Report Location	Required Confirmation / Action	Responsible Individual	Current Status
ADM-01	Declaration (Page i)	Physical signature above 'G Shailesh (23BDS0236)'	G Shailesh	**PENDING SIGNATURE**
ADM-02	Declaration (Page i)	Physical signature above 'N Neha (23BDS0270)'	N Neha	**PENDING SIGNATURE**
ADM-03	Declaration (Page i)	Physical signature above 'P Surya (23BDS0327)'	P Surya	**PENDING SIGNATURE**
ADM-04	Certificate (Page ii)	Project Guide signature & department seal	Dr. Kankana Roy (Guide)	**PENDING SIGNATURE**
ADM-05	Certificate (Page ii)	Panel In-Charge signature & name confirmation (Panel P028)	Dr. Kanagaraj K / Panel In-Charge	**PENDING REVIEW 3**

Item ID	Report Location	Required Confirmation / Action	Responsible Individual	Current Status
ADM-06	Certificate (Page ii)	Internal / External Examiner(s) signatures	Review Examiners	**PENDING REVIEW 3**
ADM-07	Certificate (Page ii)	Dean / HOD signature & SCOPE institutional seal	Dr. Jaisankar N (Dean)	**PENDING SUBMISSION**
ADM-08	Review Details	Verify Panel P028, SJT505 venue, 28 Oct 2026 @ 11:30 AM	Student Team	**CONFIRMED ACCURATE**

3. CODEBASE VERIFICATION & TECHNICAL AUDIT

Technical Subsystem	Claim in Report	Repository Ground Truth State	Audit Finding
Object Detector	End-to-End NMS-Free YOLO (2,434,188 parameters, 9.43 MB ONNX).	public/models/yolo26n.onnx verified; possesses YOLOv10-Nano dual assignment topology.	**VERIFIED ACCURATE: Avoids claiming nonexistent official 'YOLO26'.**
Multi-Target Tracking	Two-stage ByteTrack with Kalman filtering and greedy IoU matching.	Implemented in visionsense/bytetrack.py and src/tracking/bytetrack.ts.	**VERIFIED ACCURATE: Code-inspected and tested across sequential frames.**
Optical Flow Motion	Farneback dense flow (48x27 grid) in Python; regularized Lucas-Kanade in TS.	Implemented in visionsense/optical_flow.py and src/cv/denseOpticalFlow.ts.	**VERIFIED ACCURATE: Dual-path implementations documented distinctly.**
Temporal Anomaly Engine	Bi-GRU neural network with attention; heuristic surge/dispersal/bottleneck rules.	Implemented in visionsense/temporal_analysis.py with sliding window K=24.	**VERIFIED ACCURATE: Both neural model and rule-based fallback active.**

Technical Subsystem	Claim in Report	Repository Ground Truth State	Audit Finding
Crowd Safety Engine	people_count <= 20 : NORMAL, > 20 : WARNING with 3-frame debouncing & 5.0s cooldown.	Implemented in visionsense/safety_engine.py; verified via automated test suite.	**VERIFIED ACCURATE: All automated unit tests passed 100%.**
Validation Benchmark	15 curated surveillance clips (7 violence, 8 normal) with confusion matrix metrics.	Stored in public/datasets/; automated tests evaluate manifest and metrics.	**VERIFIED ACCURATE: Explicitly labeled as functional test library.**
Public Benchmark Datasets	UCSD Ped2, ShanghaiTech, and MOT17 referenced for future formal evaluation.	Physical video assets not stored in repo; specs cited from primary literature.	**VERIFIED HONEST: Transparently stated as future controlled GPU execution.**

4. HARD-COPY PRINTING & BINDING GUIDELINES

- Paper Size & Margins: Print on Standard A4 paper (210 x 297 mm) with 100% scale (no 'fit to printable area' downscaling). The left margin is formatted at 1.25 inches to provide ample binding clearance for spiral or hard-cover binding.
- Color Printing Recommendation: Color printing is strongly recommended for the 21 figures—specifically the High-Level Architecture (Figure 4.1), DCA/Temporal Subsystem (Figure 4.4), SOC Dashboard (Figure 6.1), Canvas HUD (Figure 6.3), Confusion Matrix (Figure 7.1), and Cost Comparison (Figure 7.3).
- Double-Sided vs. Single-Sided: Review university guidelines regarding single-sided vs double-sided binding. Each major chapter and front matter section starts on a fresh page break.
- Submission Deadline: Ready for immediate review and submission on 28 October 2026 at 11:30 AM in SJT505.